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- An $(\mathbf{x}^*, \mathbf{u})$ (or $\mathbf{x}^*$) which satisfies the **Karush Kuhn Tucker (KKT)** conditions is called a **KKT** point.

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Hence $[2, 1]^T$ satisfies the **KKT** condition.

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- So **KKT** condition is **not** a necessary condition for a **local minimum**, **FJ** conditions are.

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- **KKT** is **not** a **necessary condition** for a local minimum point even when all g_i 's, f are convex functions.

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- **Theorem 11:** Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a **continuously differentiable** function.

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$$\nabla f(\mathbf{x}^*) + \sum_{i \in I} u_i \nabla g_i(\mathbf{x}^*) + \sum_{j=1}^l v_j \nabla h_j(\mathbf{x}^*) = \mathbf{0}. \quad (*)$$

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 - Does there exist an $\mathbf{x}^* \in \text{Fea}(P)$ such that $G_{0, \mathbf{x}^*} \neq D_{\mathbf{x}^*}$?
 - Does there exist an FJ point which is not a KKT point? **No**

- Consider the following problem:

$$\text{Minimize } 4x_1^2 - x_2^2 + 8x_1x_2$$

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- ✓ 4 If the objective function is changed to $2x_1^2 - x_1x_2 + x_2^2 - x_2$ and if $S = \text{Fea}(P) = \{(x_1, x_2) : x_2^2 + x_1^2 \leq 1\}$ then find all optimum solutions to this problem.

- For a problem **(P)** of the form,

Minimize $f(\mathbf{x})$

subject to $g_i(\mathbf{x}) \leq 0$, for $i = 1, \dots, m$, $\mathbf{x} \in \mathbb{R}^n$,

where all the g_i 's and f are continuously differentiable throughout $\mathbb{R}^n$, and S is the feasible region of **(P)** check the correctness of the following statements with proper justification.

- 1 If $\mathbf{x}^* \in S$ is a KKT point of the above problem then $-\nabla f(\mathbf{x}^*)$ lies in the cone generated by $\nabla g_i(\mathbf{x}^*)$, $i \in I$, where I gives the indices of the binding constraints (given by g_i 's) at $\mathbf{x}^*$.

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- T 3 If $\nabla f(\mathbf{x}^*) = 0$ for some $\mathbf{x}^* \in S$ then $\mathbf{x}^*$ is a KKT point.
- 4 If $\mathbf{x}^*$ is an FJ point and there is a solution to the FJ conditions at $\mathbf{x}^*$ with $u_0 = 0$ (u_0 is the coefficient of $\nabla f(\mathbf{x}^*)$ in the FJ conditions), then $\mathbf{x}^*$ is not a KKT point.

$G_0 = \emptyset$ ←

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- If all the g_i 's and f are convex functions and $\mathbf{x}^* \in S$ is such that $F_{0,\mathbf{x}^*} \cap G_{0,\mathbf{x}^*} = \phi$, then $\mathbf{x}^*$ is a global minimum of f in S .

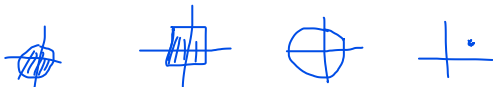
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• If $\mathbf{x}^*$ is a local minima of (P) with $F_{0,\mathbf{x}^*} \neq \phi$ but $G_{0,\mathbf{x}^*} = \phi$ then $\mathbf{x}^*$ is not a KKT point.

- There exists a (P) such that for all $\mathbf{x}^* \in S$, $G_{0,\mathbf{x}^*} = D_{\mathbf{x}^*} \neq \phi$.

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- ↑ • There exists a (P) such that $G_{0,\mathbf{x}^*} \neq D_{\mathbf{x}^*}$, for all $\mathbf{x}^* \in S$.
- ⊥ • There exists a (P) with $Fea(P) \neq \phi$ such that $G_{0,\mathbf{x}^*} = \phi$ for all $\mathbf{x}^* \in S$.
- ? • There exists a (P) with $Fea(P) \neq \phi$ such that $D_{\mathbf{x}^*} = \phi$ for all $\mathbf{x}^* \in S$.
- ? • Give examples of nonconstant functions on $\mathbb{R}^n$ which are both convex and concave and those which are neither convex nor concave.

- For a linear programming problem (P) of the form,
Minimize $\mathbf{c}^T \mathbf{x}$ subject to $A_{m \times n} \mathbf{x} \leq \mathbf{b}$, $\mathbf{x} \geq \mathbf{0}$,
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- Consider the linear programming problem.
Minimize $2x_1 - 3x_2$
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 $-(1 - x_1)^3 + x_2 = 0$
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- Consider the linear programming problem.

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$(0, 3/2)$

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Solve the KKT conditions for u_i 's. Hence find an optimal solution of the dual.

$(3/2, 7/2)$

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π_0

- Check whether $[1, 0]^T$ is a KKT point of the above problem.
How many feasible points does this problem have? What is your conclusion?